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Population-level impact of quadrivalent HPV vaccination in Brazil: a repeated cross-sectional study

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Human papillomavirus vaccination is central to cervical cancer elimination, yet population-level evidence from middle-income countries remains limited. We evaluated the impact of quadrivalent HPV vaccination in Brazil using data from two repeated population-based cross-sectional surveys conducted in 2016–2017 and 2020–2023 among sexually active individuals aged 16–25 years. Genital samples were tested for HPV genotyping and survey-weighted Poisson-regression models adjusted for sociodemographic and behavioral factors were used to compare prevalences between periods. Among 11,794 women and 4932 men, the prevalence of vaccine-type HPV declined from 15.64% (14.23–17.17) to 5.92% (5.23–6.70) among women and 13.81% (11.15–16.99) to 10.39% (8.35–12.85) among men. Adjusted analyses showed an 81% reduction among vaccinated women (PR0.19; 0.14–0.27) and a 33% reduction among unvaccinated women (PR0.67; 0.55–0.81), suggesting both direct and indirect effects. Reductions were significant only among women aged 16–23, the group with higher coverage. Among men, lower prevalence of vaccine-type HPV was observed only among vaccinated individuals (PR0.39; 0.18–0.86). The overall HPV prevalence increased modestly between the two surveys. No differences in prevalence reduction were observed according to the number of vaccine doses. These findings demonstrated the impact of HPV vaccination in Brazil and highlight the importance of maintaining high vaccination coverage and continued surveillance.

Vaccination against human papillomavirus (HPV) is one of the key prevention strategies to reduce HPV-associated cancers and achieve the World Health Organization (WHO) goals of cervical cancer elimination until 2030[1,2]. The quadrivalent HPV (4vHPV) vaccine, which protects against genotypes 6, 11, 16, and 18 (responsible for most genital warts and 80–90% of HPV-related cancers), has been incorporated into several national vaccination programs, pioneered by countries such as Australia, Mexico, and the United States as early as 2006[2].

Many real-world studies have demonstrated the vaccination impact through significant reductions in the prevalence of high-risk HPV in high-income countries. For instance, the prevalence of genotypes covered by the 4vHPV vaccine decreased from 11.5 to 3.3% when comparing the pre- and post-vaccine eras in the USA[3]. In Australia, among 18- to 24-year-old women, it dropped from 28.7 to 6.7%[4]. Moreover, the prevalence of oncogenic genital HPV infections (types 16 and 18) decreased significantly among girls and women after vaccine implementation, with greater reductions observed at younger ages, as shown by a meta-analysis with data from 14 high-income countries[5].

In Brazil, the quadrivalent vaccine was introduced in the public immunization program as a two-dose schedule in 2014, initially targeting girls aged 11–13 years. A year later, the program was expanded to include younger girls, lowering the minimum age to nine years. In 2017, it was further extended to older girls, raising the maximum age to 14 years and also including boys (aged 11–14 years) and high-risk adult populations, such as people living with HIV, oncology patients, and those who had undergone solid organ or bone marrow transplants[6].

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Since 2024, the vaccination schedule has been updated to a single dose for boys and girls aged 9–14 years, and for the catch-up of unvaccinated adolescents of both sexes aged 15–19 years in 2025. The HPV vaccination was also expanded to pre-exposure prophylaxis users, victims of sexual violence, individuals with recurrent respiratory papillomatosis[6]. Considering the first dose, the vaccination coverage in Brazil was estimated at 83.21% among women and 67.57% among men aged 9–14 years in 2024[7].

Just three years after the vaccine's introduction in Brazil, our previous findings showed that vaccinated girls had significantly lower prevalence of genotypes covered by the 4vHPV vaccine compared to unvaccinated ones[8]. However, a decade after the implementation of the quadrivalent HPV vaccine in Brazil, the real population-level impact on HPV prevalence remains largely unknown when comparing pre- and post-vaccine eras. While several studies have demonstrated positive vaccine impact in HPV prevalence, they primarily focus on high-income countries, with limited data from Latin America, a region with extensive vaccination programs and high cervical cancer incidence[9,10]. Therefore, we aimed to evaluate the population-level impact of HPV vaccination on HPV prevalence among Brazilians aged 16–25.

Results

Characteristics of studied participants

A total of 6388 participants (1120 men and 5268 women) from POP-Brazil 1 and 10,338 (3812 men and 6526 women) from POP-Brazil 2 were analyzed (Fig. 1). Some differences in sample characteristics between the two surveys were noted. The median (IQR) age of participants slightly decreased from POP-Brazil 1 to POP-Brazil 2 (21.7;19.5–23.8 versus 21.5; 19.3–23.6), mainly explained by the higher proportion of younger women (aged 16–17 years) in POP-Brazil 2. Although the proportion of white participants remained stable, the distribution of black and pardo participants shifted among both men and women, with a higher proportion of black and a lower proportion of pardo participants in POP-Brazil 2. Among women, the proportion of participants in socioeconomic class A-B increased from 12.7% (11.4–14.0) to 16.0% (14.7–17.3). The use of hormonal contraceptives also increased over time (from 48.1%; 46.0–50.1 to 68.4%; 66.5–70.3) (Table 1).

The proportions of participants not in a relationship and with multiple sexual partners in the last year increased significantly among both men and women. Among men, the proportions of individuals who initiated sexual activity after age 15 increased from 59.9% (55.0–64.7) to 66.3% (62.7–69.7). No differences were observed in alcohol consumption or reported condom use at last intercourse. In POP-Brazil 2, there were 116 men (3.6%) and 32 women (0.6%) living with HIV, of whom 44 men and 12 women had received HPV vaccination. There were no differences in the HIV prevalence among the two surveys. Vaccination coverage in POP-Brazil 2 was significantly lower among men (31.1%; 27.4–35.0) compared to women (61.8%; 60.1–63.5) (Table 1). It is important to consider that POP2 participants (aged 16–25 years, birth cohorts 1995–2007) faced substantially different vaccination opportunities due to changes in the HPV vaccination policies through time (Supplementary Fig. 1)[6]. Older participants depended more on private access, and many had already exceeded the recommended age range by 2014, making vaccination less likely, particularly among boys. The weighted mean time since vaccination was 8 years (7.8–8.1) among women and 6.5 years (5.7–7.2) among men.

Prevalence of HPV across surveys and impact of vaccination

The overall HPV prevalence in POP-Brazil 1 and POP-Brazil 2 was 46.7% (44.5–48.9) and 56.6% (54.6–58.5), respectively, showing a statistically significant increase between the surveys (Fig. 2). In the adjusted model, this increase remained statistically significant for women, but not men (Supplementary Table 1). However, the prevalence of genotypes covered by the quadrivalent vaccine declined from 14.98% (13.61–16.45) in POP-Brazil 1 to 7.95% (6.92–9.12) in POP-Brazil 2, irrespective of vaccination status or sex (Supplementary Table 2). To describe vaccine impact at a population-level, we compared all participants in the POP-Brazil 2 to all participants from POP-Brazil 1 leading to an adjusted PR of 0.53 (0.44–0.64). Compared to the first survey, the adjusted prevalence ratio of 4vHPV genotypes decreased by 23% (PR 0.77; 0.63–0.94) among unvaccinated participants

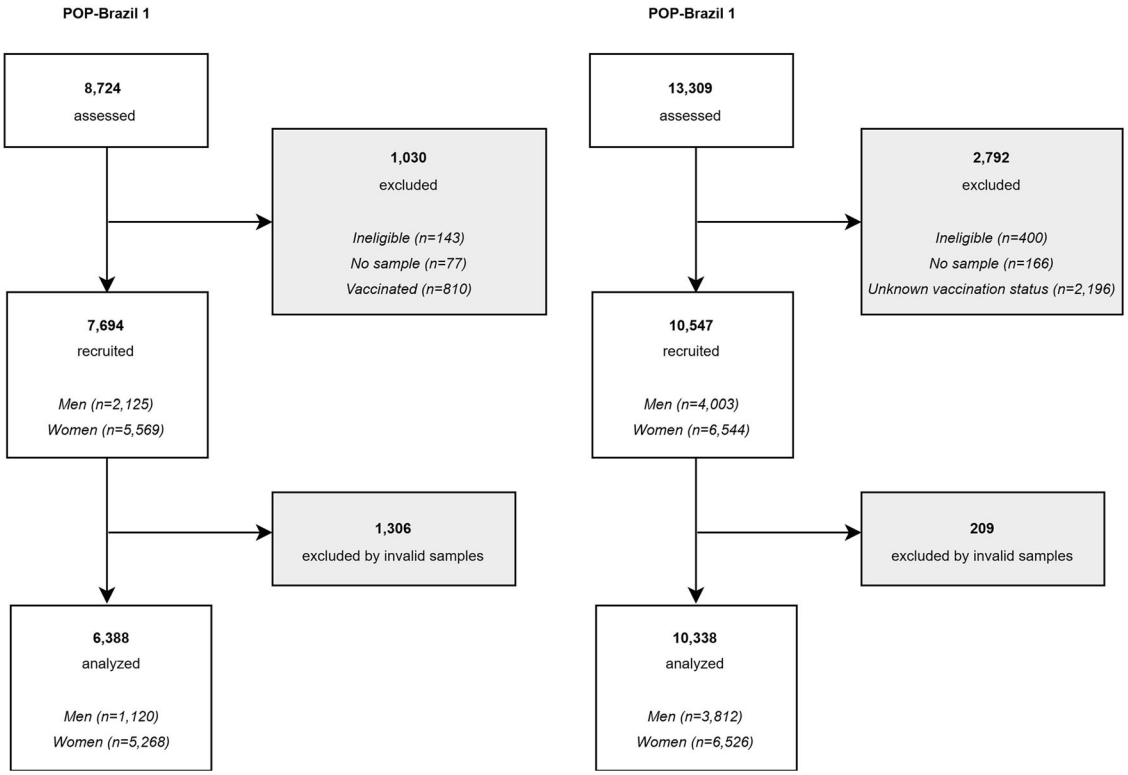


Fig. 1 | Flowchart of participant inclusion in the POP-Brazil 1 and POP-Brazil 2.

Table 1 | Characteristics of the POP1 and POP2 participants, stratified by sex

	POP1 Men		POP2 Men			POP1 Women		POP2 Women		
Variable	**n**	**% (95% CI)**	**n**	**% (95% CI)**	**p value[a]**	**n**	**% (95% CI)**	**n**	**% (95% CI)**	**p value[a]**
Age					0.6					<0.001
16–17	124	13.9 (10.7–17.7)	352	10.5 (8.39–13.1)		429	7.7 (6.73–8.73)	842	14.1 (12.8–15.5)	
18-19	279	19.4 (15.9-23.5)	929	19.7 (17.3–22.4)		1107	21.7 (20.0–23.4)	1250	19.9 (18.5–21.5)	
20–21	274	23.4 (19.6–27.7)	793	24.5 (21.1–28.1)		1209	22.0 (20.4–23.7)	1455	22.0 (20.5–23.6)	
22-23	232	21.9 (18.2–26.1)	912	22.9 (20.2–25.9)		1233	23.8 (22.1–25.6)	1604	24.5 (23.0-26.1)	
24–25	211	21.4 (17.7–25.6)	826	22.4 (19.5–25.6)		1290	24.8 (23.1–26.7)	1375	19.4 (18.1–20.8)	
Skin color/Ethnicity					0.027					<0.001
White	266	24.9 (20.9–29.4)	1122	25.5 (22.6–28.6)		1253	22.4 (20.6-24.2)	1574	23.7 (22.3–25.3)	
Black	198	18.9 (15.4–23.0)	771	25.9 (22.5–29.6)		815	16.8 (15.3–18.3)	1215	21.5 (19.9–23.1)	
Pardo	626	54.8 (49.9–59.6)	1863	47.6 (44.0–51.2)		3045	58.2 (56.1–60.2)	3607	53.0 (51.2–54.9)	
Other	17	1.4 (0.605–3.04)	56	1.0 (0.678–1.55)		134	2.7 (2.03–3.47)	127	1.8 (1.35–2.28)	
Socioeconomic class					0.6					<0.001
A-B	323	23.3 (19.7–27.5)	1079	26.1 (23.0–29.4)		765	12.7 (11.4–14.0)	1086	16.0 (14.7–17.3)	
C	612	58.3 (53.5–63.0)	2153	56.9 (53.3–60.5)		2790	53.8 (51.8–55.9)	3588	56.2 (54.4–58.0)	
D-E	185	18.3 (14.9-22.4)	579	17.0 (14.5–19.8)		1713	33.5 (31.6–35.4)	1851	27.8 (26.2–29.5)	
Alcohol consumption	826	73.7 (69.1–77.8)	2391	69.1 (65.9–72.2)	0.10	3247	60.4 (58.3–62.4)	3972	62.8 (61.0–64.6)	0.071
Partnership status					<0.001					<0.001
Single/Widowed/Divorced/ Separated	376	31.4 (27.0–36.2)	1974	48.9 (45.2–52.5)		1038	19.1 (17.6–20.8)	1777	28.0 (26.4–29.7)	
Dating	474	44.3 (39.6–49.1)	1276	35.3 (31.9-38.7)		1929	37.4 (35.4–39.3)	2608	41.7 (39.9–43.5)	
Married/Living together	270	24.3 (20.5–28.5)	562	15.9 (13.6-18.5)		2300	43.5 (41.5–45.6)	2140	30.3 (28.7–32.0)	
Condom use (last intercourse)	527	41.9 (37.3–46.6)	1521	41.9 (38.3-45.6)	>0.9	1882	35.2 (33.3–37.2)	2192	32.8 (31.2–34.6)	0.073
Sexual partners (last year)					0.048					<0.001
Up to one	523	52.8 (47.7–57.8)	1583	46.4 (42.7–50.2)		3986	78.9 (77.2–80.5)	4407	70.1 (68.3–71.8)	
Two or more	496	47.2 (42.2–52.3)	1759	53.6 (49.8–57.3)		1140	21.1 (19.5–22.8)	1842	29.9 (28.2–31.7)	
Age at sexual debut					0.036					>0.9
<15	423	40.1 (35.3-45.0)	1091	33.7 (30.3-37.3)		1606	31.6 (29.7–33.5)	1987	31.7 (30.0–33.4)	
≥15	632	59.9 (55.0–64.7)	2538	66.3 (62.7–69.7)		3582	68.4 (66.5–70.3)	4484	68.3 (66.6–70.0)	
Hormonal contraceptive use	–	–	–	–		2598	48.1 (46.0–50.1)	3216	68.4 (66.5–70.3)	<0.001
HIV					0.3					>0.9
Negative	583	97.8 (94.9–99.1)	2912	96.4 (94.9–97.5)		3681	99.4 (98.9–99.7)	5587	99.4 (99.1–99.7)	
Positive	13	2.2 (0.92–5.1)	116	3.6 (2.47–5.13)		20	0.6 (0.311–1.08)	32	0.6 (0.34–0.94)	
HPV vaccine coverage					–					-
Vaccinated	0	0	763	31.1 (27.4–35.0)		0	0	3643	61.8 (60.1–63.5)	
Unvaccinated	1120	100	3049	68.9 (65.0–72.6)		5268	100	2883	38.2 (36.5–39.9)	

[a]Pearson's X^2: Rao & Scott adjustment.

and by 74% (PR 0.26; 0.18–0.37) among vaccinated participants in POP-Brazil 2 (Table 2). Some factors are independently associated to HPV prevalence in both survey as alcohol consumption and number of sexual partners (Supplementary Table 3). We further conducted a stratified analyses to better evaluate the impact of age on HPV prevalence. The decrease in HPV prevalence is visible only on the youngest group, in both women and men, but not in the older ones (Supplementary Table 4).

Vaccination was associated with lower prevalence of genotypes covered by the 4vHPV vaccine in both sexes. The overall prevalence of HPV increased in both unvaccinated women and men between surveys, whereas high-risk HPV increased only among women (Fig. 2). The increase in HPV prevalence among women was mainly due to the rise in high-risk HPV types 35, 39, 51, 56, 58, and 68, as well as low-risk HPV types 40, 42, 54, 70, and 82. We observed over 80% reductions in the prevalence of HPV types 6, 16, and 18 when comparing women from POP-Brazil 1 with vaccinated women in POP-Brazil 2. There was a non-significant prevalence reduction for HPV-11 (Supplementary Table 5).

The prevalence of HPV types covered by the 4vHPV vaccine decreased from 15.6% (14.2–17.2) in POP-Brazil 1 to 10.5% (9.18–12.1) among unvaccinated women in POP-Brazil 2. A substantially greater reduction was observed among vaccinated women, in whom the prevalence reached 3.1% (2.38–3.95) (Fig. 2). These results remain significant also in the multivariate analysis, with a PR of 0.19 (0.14–0.27) (Table 2). Age-stratified analysis indicated that this decrease was most pronounced in the age group with highest vaccine coverage, who exhibited markedly lower prevalence of 4vHPV types compared with their counterparts in the first survey (Fig. 3). However, stratified analysis also showed that vaccinated women 24–25 years did not experience the same reduction in HPV prevalence multivariate analysis; PR 0.64 (0.39–1.05) (Supplementary Table 4). Evidence of cross-protection was also observed for phylogenetically related HPV types. In

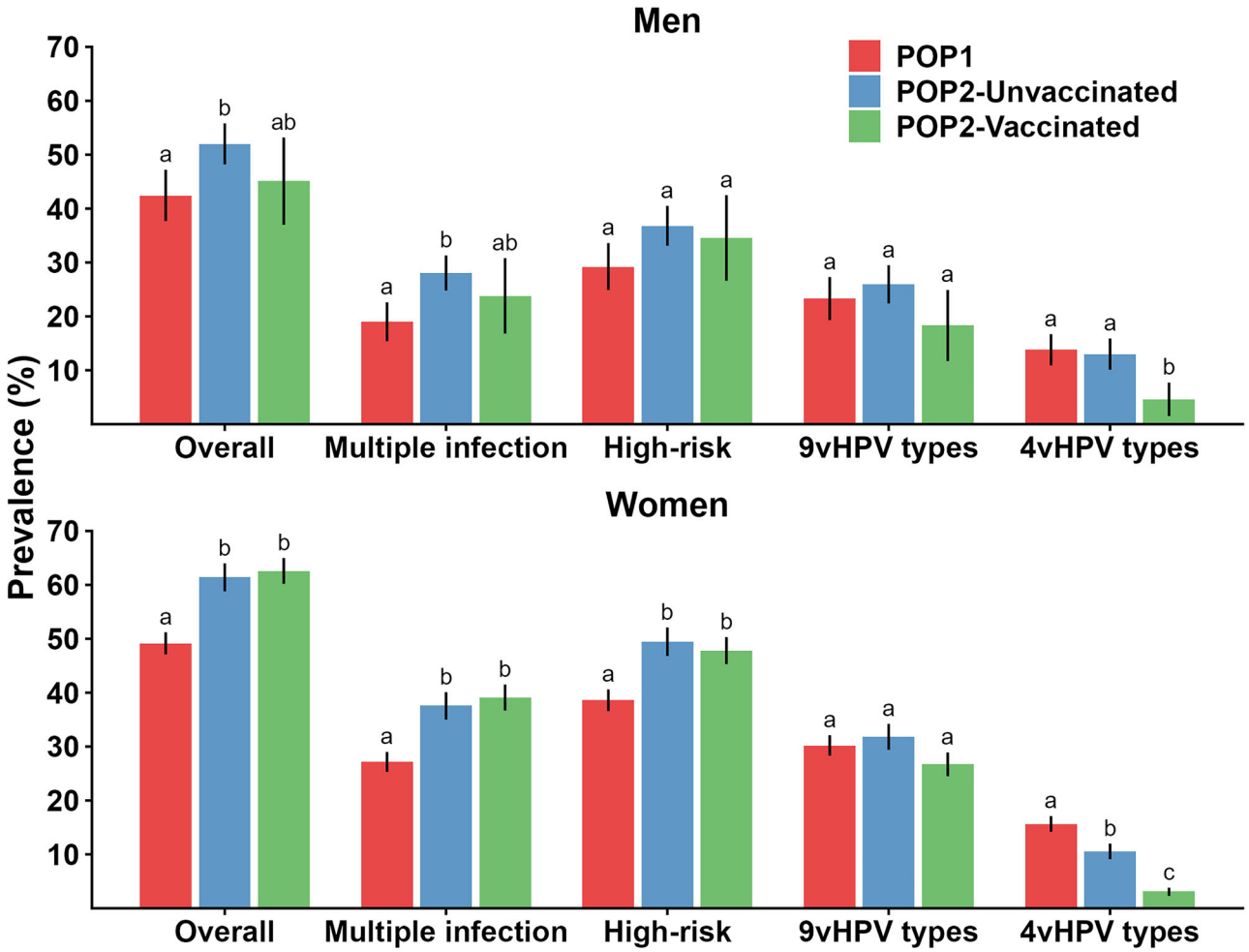


Fig. 2 | Genital HPV prevalence among POP-Brazil 1 and POP-Brazil 2 participants, stratified by vaccination status and sex. Prevalences estimates are weighted by age and sex. Vertical lines represent 95% confidence intervals. Different letters above the bars (**a**–**c**) indicate non-overlapping confidence intervals, suggesting a statistically significant difference between groups.

Table 2 | Genital 4vHPV genotypes prevalence ratio estimates for POP-Brazil 1 and POP-Brazil 2

	Crude		Adjusted	
Group	**PR**	**95% CI**	**PR**	**95% CI**
Overall (n = 16,726)				
POP1	1.00	—	1.00	—
POP2-Unvaccinated and vaccinated	0.53	(0.45–0.63)	0.53[a]	(0.44–0.64)
POP2-Unvaccinated	0.80	(0.67–0.96)	0.77[a]	(0.63–0.94)
POP2-Vaccinated	0.24	(0.17–0.32)	0.26[a]	(0.18–0.37)
Men (n = 4932)				
POP1	1.00	—	1.00	—
POP2-Unvaccinated	0.94	(0.69-1.28)	0.88[b]	(0.62–1.27)
POP2-Vaccinated	0.33	(0.17–0.67)	0.39[b]	(0.18–0.86)
Women (n = 11,794)				
POP1	1.00	—	1.00	—
POP2-Unvaccinated	0.67	(0.57–0.79)	0.67[b]	(0.55–0.81)
POP2-Vaccinated	0.20	(0.15–0.26)	0.19[b]	(0.14–0.27)

[a]Overall adjusted for sex, age group, alcohol consumption, number of sexual partners, and HIV status.

[b]Sex-stratified analyses adjusted for age group, alcohol consumption, number of sexual partners, and HIV status.

particular, vaccinated women aged 16–17 showed a 62% reduction in HPV31 prevalence compared with unvaccinated women in POP-Brazil 1 (PR 0.38; 0.17–0.83). Conversely, overall HPV prevalence increased only among women aged 22 or older, whereas the prevalence of high-risk HPV and multiple HPV infections increased across all women, regardless of age group or vaccination status (data not shown).

Among men, the prevalence of 4vHPV types declined from 13.8% (11.2–17.0) in POP-Brazil 1 to 4.6% (2.35–8.88) among vaccinated men in POP-Brazil 2 (Fig. 2). There was a 61% reduction in the prevalence of HPV also in the multivariate analysis (PR 0.39; 0.18–0.86) (Table 2). When stratified by age, the reductions were significant only among those aged 16–22 years old (PR 0.21; 0.10-0.41) (Supplementary Table 4).

Vaccine effectiveness and doses of HPV vaccine

We also compared the prevalence of HPV contained in the quadrivalent vaccine among participants of the POP-Brazil 2 to estimate the effectiveness of vaccination. Comparing vaccinated and unvaccinated participants in the multivariate analysis, there was a significant 53% reduction in 4vHPV prevalence (PR 0.47; 0.32–0.71) (Supplementary Table 3). We additionally evaluated the prevalence of HPV according to the number of doses received. There were no statistical differences in the prevalence of HPV types covered by the quadrivalent vaccine among people who received one, two, or three doses. Similar results were observed for both men and women (Table 3).

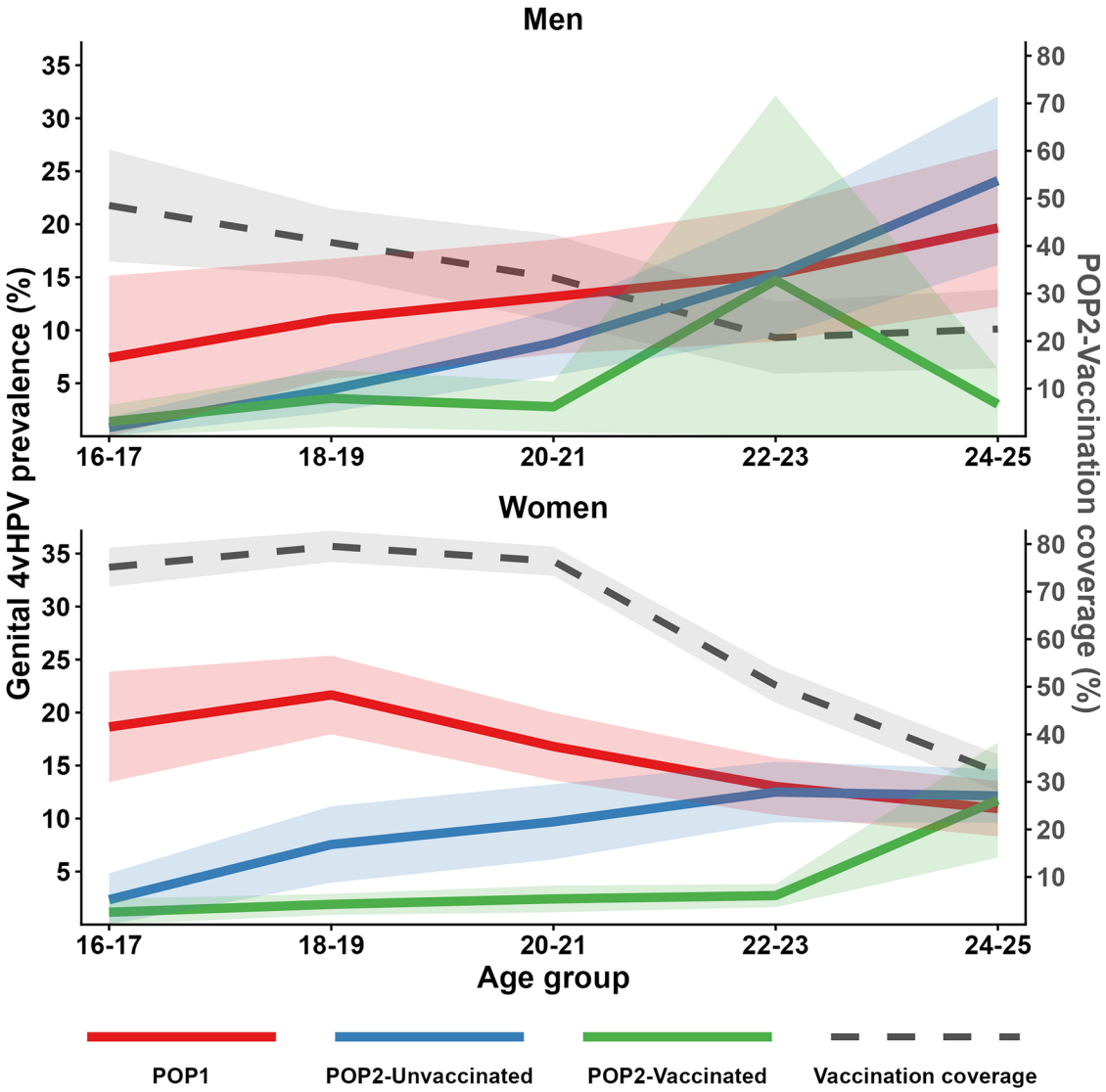


Fig. 3 | Genital prevalence of the genotypes covered by the 4vHPV vaccine by age groups among POP-Brazil 1 and POP-Brazil 2 participants, stratified by vaccination status and sex. Solid lines represent prevalence estimates weighted by age and sex. Dashed lines represent the weighted proportion of vaccinated participants from POP-Brazil 2 study. Shaded areas around lines show 95% confidence intervals.

Table 3 | Genital prevalence of 4vHPV vaccine types among POP-Brazil 2 participants, by sex and vaccination doses

	Overall		Men		Women	
HPV Vaccination	Cases	% (95% CI)	Cases	% (95% CI)	Cases	% (95% CI)
Unvaccinated	687	12.00 (10.30–13.95)	318	12.99 (10.37–16.14)	369	10.53 (9.18–12.05)
Vaccinated	182	3.53 (2.61–4.76)	46	4.62 (2.35–8.88)	136	3.07 (2.38–3.95)
Vaccinated (1 dose)	26	2.53 (1.53–4.16)	9	2.86 (1.15–6.94)	17	2.31 (1.34–3.96)
Vaccinated (2 doses)	86	2.55 (1.84–3.51)	20	3.36 (1.78–6.27)	66	2.28 (1.56–3.31)
Vaccinated (3 doses)	21	2.78 (1.67–4.59)	5	5.67 (2.16–14.08)	16	2.43 (1.34–4.38)
Vaccinated (Unsure)	49	7.24 (3.84–13.22)	12	7.35 (2.06–23.01)	37	7.15 (4.41–11.39)

Discussion

This was the largest study to assess the impact of the quadrivalent HPV vaccine in Latin America. There was a clear decline in the prevalence of HPV types covered by the 4vHPV vaccine among young people. A decline was observed not only among vaccinated individuals but also at the population level, including unvaccinated women. These findings indicate that the national HPV immunization program effectively prevents the spread of oncogenic HPV genotypes included in the quadrivalent vaccine among sexually active women and men after the introduction of the vaccine in Brazil in 2014. Vaccine coverage plays a pivotal role in controlling HPV

transmission. The reduction in HPV prevalence was more pronounced in age groups with higher vaccine coverage, as demonstrated by the age-stratified analysis. There was no difference in HPV prevalence according to the number of doses of vaccine.

Findings from POP-Brazil had already demonstrated the vaccine's effectiveness in Brazil, with prevalence of HPV types 6, 11, 16, and 18 declining by 56.8% (from 15.64% in unvaccinated women to 6.76% in vaccinated women)[8]. As a repeated cross-sectional survey, POP-Brazil 2 demonstrated substantial vaccine impact: between the surveys, multivariate analysis showed that vaccination led to an 81% reduction in the vaccine-targeted genotypes among vaccinated women and a 61% reduction among vaccinated men. Notably, younger women showed a greater reduction in the prevalence of HPV genotypes covered by the 4vHPV vaccine compared to older women.

These findings are consistent with results from HPV vaccination programs throughout the Americas, spanning high-, middle-, and low-income countries. In high-income countries, vaccine impact ranged from 74 to 93% in adolescents aged 9–14 years and 12–90% in those aged 15–18[11]. A meta-analysis from high-income countries similarly demonstrated higher reductions among younger girls, with HPV 16 and 18 prevalence decreasing by 83% among those aged 13–19 compared to 66% among women aged 20–24 years, after 5–8 years of vaccine implementation[5]. In Colombia, data from individuals in the same age range show a reduction in HPV 16 and 18 from 15.4 to 6.5%, with higher impact for vaccination occurring before sexual debut[12]. Similarly, vaccination decreased the prevalence of overall and quadrivalent-specific HPV types among girls recruited in public hospitals in Argentina[13]. In the United States, data from New Mexico show a 75–80% relative reduction in prevalence of HPV genotypes covered by the 4vHPV vaccine among screened women, while the National Health and Nutrition Examination Survey (NHANES) demonstrates an even greater decrease, from 18.5% pre-vaccination to 2.8% after vaccine implementation[3,14]. Beyond demonstrating clinical impact, evidence from a systematic review indicates that vaccinating girls against HPV is a cost-effective strategy for Latin American countries[15].

We also observed a reduction in the prevalence of genotypes covered by the 4vHPV vaccine among non-vaccinated younger women (under 21 years), indicating an indirect population-level effect of vaccination on HPV transmission within groups with high vaccination coverage. Herd protection effects among unvaccinated women have been documented internationally, with reductions in vaccine-type HPV prevalence of 35% in Australia (6 years post-vaccination)[16], 55% in the United States (9–10 years post-vaccination)[17], and 87% reduction in HPV 16/18 odds in Scotland (7 years post-vaccination)[18]. In contrast to our findings, similar effects have also been suggested among men in Australia[19], Canada[20], and Denmark[21]. This difference is likely explained by the substantially higher vaccination coverage in these countries compared to Brazil, as predictive modeling suggests herd protection magnitude increases proportionally with vaccine uptake[22]. Beyond herd effects, data from the Netherlands showed cross-protection against the non-targeted HPV31 type in bivalent-vaccinated women[23]. Randomized clinical trials and observational studies suggest that cross-protection is largely driven by HPV types 31 and 45[24]. We also observed decreased HPV-31 prevalence among younger vaccinated women, supporting cross-protection and consistent with findings of reduced HPV-31/33 prevalence in women aged 21–25 years[14].

Regarding non-vaccine HPV types, our results suggested no overall reduction in HPV prevalence or even increasing trends among women. Post-vaccination trends in non-quadrivalent HPV types have been heterogeneous and required careful interpretation. While declines in vaccine-targeted genotypes are well established, some studies report reductions in non-targeted types[4,25], whereas others describe increasing trends in non-vaccine HPV types[14,17,19,21]. For example, recent studies from the Netherlands and New Mexico have reported increases in several high-risk types, including HPV 39, 45, 51, 52, 56, and 68[14,23]. Similarly, HPV types 39, 51, 56, and 68 increased from POP-Brazil 1 to POP-Brazil 2. However, these patterns might not reflect the impact of vaccination. Changes in sexual behaviors of young people, particularly the increased number of sexual partners may contribute to these trends[5,26]. Another possible explanation is type replacement, whereby non-vaccine HPV types expand to occupy the ecological niche left by the decline of HPV-16/18[5,27]. Therefore, these findings should be interpreted with caution given the current uncertainty in the literature.

The decrease in prevalence of genotypes covered by the 4vHPV vaccine was higher in women than in men, particularly among individuals younger than 21 years. Differences in HPV vaccination uptake opportunities should be considered when interpreting vaccine impact and coverage by age group and sex. Most older participants (≥ 21 years) had already exceeded the initial target age at vaccine introduction in 2014 and were therefore less likely to have been vaccinated. This was particularly true for boys, who were only included in the program from 2017 onward[6]. As a result, reductions in HPV prevalence among men in general are expected to rely largely on indirect protection conferred by high vaccination coverage in women, which may explain the less pronounced decrease observed in men in POP2. This finding is not unexpected, as boys typically receive less routine support for HPV vaccination than girls, partly because HPV is still widely perceived primarily as a cause of cervical cancer. Consequently, most studies evaluating the impact of HPV vaccination have focused exclusively on girls. Among the 164 countries with HPV vaccination programs monitored by the WHO, only 85 (51.8%) have expanded vaccination for boys until 2026[28]. A systematic review showed that gender-neutral HPV vaccination is a cost-effective strategy, although its cost-effectiveness is highly dependent on vaccination coverage[29].

There was no difference in the prevalence of 4vHPV types according to the number of vaccine doses received, suggesting that a single-dose HPV vaccination regime may be sufficient to confer protection, in line with current WHO recommendations[1]. Consistent with these findings, a recent systematic review found no evidence of differences in HPV16 and HPV18 detection across different dose schedules among vaccinated individuals[30].

Some limitations should be noted. Only participants from capital cities were included, who may have different sexual practices and behaviors compared to those living in non-capital or rural areas. Vaccine status was assessed by both vaccination record and self-report. Although a sensitivity analysis restricted only for those with registered vaccination dates did not substantially change the results, misclassification bias could not be ruled out. The interval between vaccine doses was also not assessed. Additionally, the genotyping method used in the first round of POP-Brazil was discontinued, necessitating a change in methodology for the second round (Anyplex™ II HPV28 Detection, replacing the Linear Array® Genotyping Test). We performed a cross-validation analysis to compare the two methods[31]. This analysis revealed that while the detection rates for all analyzed HPV types were not significantly different, HPV-6 specifically demonstrated a lower Kappa agreement and lower sensitivity using the Anyplex™ II method. Consequently, the potential impact of this methodological difference on the observed decrease in HPV-6 prevalence in the second round of POP-Brazil cannot be ruled out. To mitigate further impacts of the different genotype panels we analyzed only the 26 HPV genotypes which were present in both methods. Nonetheless, the study has several strengths. Both surveys employed comparable methodologies across distinct time periods (pre- and post-HPV vaccine implementation) and collected detailed information on sexual behavior and other risk factors. This design provides a robust and well-powered framework for assessing temporal trends and population-level changes associated with vaccination and HPV-related risk factors. In addition, weighted prevalence estimations improved comparability across surveys by accounting for unequal representation across capitals and between sexes, while multivariate analysis controlled for potential confounding factors when estimating the effect of the vaccination on genital HPV prevalence.

The evidence from this large national study demonstrates the substantial impact of the quadrivalent HPV vaccination program in reducing HPV transmission among sexually active women and men 5–9 years after its implementation. The consistent decline in the prevalence of genotypes

covered by the 4vHPV vaccine, observed not only among vaccinated participants but also across the unvaccinated women, highlights the effect of widespread vaccination in reducing HPV transmission and protecting future generations against cancer and other HPV-related diseases.

Methods

Study design/population

We used data from repeated cross-sectional surveys to compare genital HPV prevalence among sexually active 16- to 25-year-old men and women between pre- and post-vaccine eras in Brazil. The first study (POP-Brazil 1) was conducted between 2016 and 2017, in the 26 state capitals and the Federal District of Brazil, right after the introduction of the HPV vaccination program. The second study (POP-Brazil 2), was conducted between 2020 and 2023 with the same population target. Participants who had not initiated sexual activity, pregnant women, women who had given birth within the previous six months, women who had undergone a hysterectomy or trachelectomy, and women who had ever had grade 2 or higher cervical intraepithelial neoplasia were not included. Both phases of the POP-Brazil survey were conducted according to the original study protocol[11], ensuring consistent methodological characteristics and data collection procedures that enabled data comparison across the two time periods.

Participants were recruited from public health units and responded to a structured questionnaire conducted by health professionals who received rigorous training and onsite certification. Vaccination status was assessed as a dichotomous variable (vaccinated/unvaccinated). Participants were classified as vaccinated against HPV if they self-reported vaccination or had documented vaccination history (verified at the time of participation or through medical records and vaccination registries), regardless of dose number and vaccine type. Time since vaccination (years) was calculated by subtracting self-reported age at vaccination from age at interview. Sociodemographic variables included age, self-reported skin color/ethnicity, and socioeconomic status, which was computed based on a score of self-reported number of household goods according with the Associação Brasileira de Empresas de Pesquisa (ABEP). We also explored alcohol consumption in the last year, partnership status, condom use during the last sexual intercourse, number of sexual partners in the last year, and the age of sexual debut. Because retesting for HIV is not recommended for individuals who are already confirmed to be HIV-positive, we first assessed HIV status. For those that are HIV negative, unsure, or never have been tested, an HIV test was offered, with the result being incorporated in the HIV status determination.

Sample collection

Sample collection followed the study protocol[32]. Cervical samples were collected by a health professional, while penile samples were obtained through supervised self-collection using one sterile swab scrubbed across the penile shaft, glans penis, and coronal sulcus. Samples were stored at controlled room temperature (15-25 °C) and shipped to the laboratory of clinical epidemiology at the Federal University of Health Sciences of Porto Alegre (EpiClin-UFCSPA), where they were aliquoted and kept at -80 °C until use.

DNA extraction and HPV genotyping

Viral total nucleic acids were extracted from genital samples using either the LC DNA Isolation Kit III on the MagNA Pure LC 2.0 instrument (Roche) or the Maxwell® RSC Viral Total Nucleic Acid Purification Kit All on a Maxwell® RSC 48 instrument (Promega), following manufacturer's protocols. In POP-Brazil 1, HPV genotyping was performed by hybridization assay using a Linear Array® Genotyping Test (Roche) on the AutoBlot instrument (Fujirebio). Type-specific real-time PCR was used when needed to resolve cross-reactivity between HPV52 with types 33, 35, and 58. In POP-Brazil 2, HPV detection and genotyping was carried out using the Anyplex™ II HPV28 Detection (Seegene), a multiplex PCR which simultaneously detects and quantifies 28 distinct HPV types. Real-time PCR reactions were performed according to the manufacturer's instructions using 5 µL of DNA and were performed on a CFX96™ instrument (Bio-Rad), following the manufacturer's protocols, with automated data interpretation using Seegene Viewer software (v2.0). Both genotyping methods show comparable performance in clinical specimens, with an overall agreement of 96%, though the genotype panels are not identical[31].

To ensure comparability of genital HPV prevalence between POP-Brazil 1 and POP-Brazil 2, prevalence was estimated based on positive cases for the 26 genotypes detected by both the Linear Array® Genotyping Test (used in POP1) and the Anyplex™ II HPV28 Detection (used in POP2): 6, 11, 16, 18, 26, 31, 33, 35, 39, 40, 42, 45, 51, 52, 53, 54, 56, 58, 59, 61, 66, 68, 69, 70, 73, and 82. Results were categorized as: (i) overall HPV (positive for any genital HPV); (ii) multiple infection (more than one genital HPV type detected); (iii) high-risk HPV (positive for any genital high-risk HPV type: 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, and 68), whose definition was based on the test classification derived from the International Agency for Research on Cancer, according to study protocol; (iv) high-risk genital HPV except 16 and 18; (v) 4vHPV (HPV types included in the quadrivalent vaccine: 6, 11, 16, 18); and (vi) 9vHPV (types in the nonavalent vaccine: 6, 11, 16, 18, 31, 33, 45, 52, and 58). Results were considered negative when none of the 26 overlapping HPV genotypes were detected. Invalid (missing) samples due to insufficient genetic material for analysis were excluded (Fig. 1).

Statistical analysis

Both surveys used a convenience sample in each capital, with sample sizes structured to reflect the regional population distribution of the Brazilian Census. Although convenience sampling lacks defined selection probabilities, individual-level weights were calculated under the assumption that selection probabilities within each sex-capital stratum were proportional to their representation in the target population aged 16–25 years. To improve comparability across surveys, while accounting for unequal representation across capitals and between sexes, weights were defined as the inverse of the sampling fraction (the ratio of the stratum sample size to the population aged 16–25 years in that stratum), as recorded in the 2010 Census data from the Brazilian Institute of Geography and Statistics. All analyses were performed using R Statistical Software (v4.5.3; R Core Team 2026) and weights were incorporated using the 'srvyr' R package (v1.3.0). Statistical significance was set at $\alpha = 0.05$ with 95% confidence intervals. Characteristics of the participants, stratified by sex, were compared using Pearson's $\chi 2$ tests with Rao & Scott second-order corrections. Survey-weighted Wilcoxon rank-sum test was used for age comparisons. For the comparison of the prevalence ratio (PR) of HPV genotypes between surveys, Poisson regression models with robust variance were employed through the 'svyglm' function of the 'survey' R package (v4.4-8) with 'quasipoisson(link = "log")' as the family argument. Final models were adjusted for sex, age group, alcohol consumption, number of partners (last year), and HIV status.

Ethics

All participants provided written informed consent. Confidentiality was ensured during data collection and the publication of the results.

POP-Brazil 1 and POP-Brazil 2 were approved by the Ethics Committee of Moinhos de Vento Hospital (approval numbers 1,607,032 and 4,129,472, respectively), as well as by the Ethics Committees of all participating centers: Centro de Ciências da Saúde da Universidade Federal da Paraíba (28696320.9.3001.5188), Escola de Saúde Pública da Secretaria de Saúde do Rio Grande do Sul (93485418.7.3014.5312), Fundação Antônio Prudente - A.C. Camargo Cancer Center (93485418.7.3002.5432), Fundação de Ensino e Pesquisa em Ciências da Saúde do Distrito Federal (28696320.9.3064.5653), Fundação Escola de Saúde Pública de Palmas (28696320.9.3018.9187), Hospital Distrital Gonzaga Mota Messejana (28696320.9.3002.8145), Hospital e Maternidade Dona Iris (95320718.4.3003.5058), Hospital Nossa Senhora da Conceição (93485418.7.3003.5530), Instituto de Gestão Estratégica de Saúde do Distrito Federal (93485418.7.3009.8153), Instituto de Medicina Integral Professor Fernando Figueira (93485418.7.3018.5201), Irmandade da Santa

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to ongoing analyses for additional publications and because they contain sensitive information, but are available from the corresponding author on reasonable request.

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Author contributions

EMW: conceptualization, data curation, funding acquisition, investigation, methodology, project administration, resources, supervision, and writing (original draft). RSM, NLK, GE and TMD: investigation, methodology, and writing (original draft), with GE also responsible for formal analysis and data curation, AGKM, PCG, LV, DBCN and ASB: conceptualization and writing (review & editing). All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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